



Course Syllabus * SPRING 2024

INST 327 – Database Design and Modeling

Sections: 010x, 020x, and ESG1

Course Description and Learning Outcomes

A broad introduction to relational database systems, this course will provide students with a combination of conceptual understanding and technical practice. Students will learn about the relational model, which provides the logical framework for designing and querying relational databases. Students will also learn important technical and conceptual approaches to database design, including user-oriented design, requirements analysis and specification, entity relationship modeling, and normalization. Students will put these fundamentals into action by learning and using the Structured Query Language (SQL) and a database management system (DBMS) to build, populate, and query a working database.

After successfully completing the course, you will be able to:

- Describe the relational model as a logical system for structuring data for retrieval.
- Create user-oriented database queries using the Structured Query Language (SQL).
- Translate user needs into functional database requirements by using entity-relationship models that conform to the relational model.
- Build a working relational database using a database management system (DBMS).
- Normalize and denormalize a relational database to optimize performance.
- Identify security issues in databases and develop approaches to address them.
- Explain how the use and design of relational database systems reflect broader social and organizational structures and the related ethical and equity issues.

Instructors

Dr. Vedat Diker, [pronouns: he/him](#)

Prof. Pamela Duffy, [pronouns: she/her](#)

In-Class Meetings Lecture:

010x	W	2:00-2:50pm	TMH 0301
020x	Th	11:00-11:50am	ESJ 2204
ESG1	F	10:00-10:45am	BLD4 4335

Discussion/Lab Sessions:

0101	F	9:00-9:50am	CCC 1111
0102	F	10:00-10:50am	ATL 2428
0103	F	11:00-11:50am	ATL 1113
0201	F	2:00-2:50pm	PLS 1130
0202	F	3:00-3:50pm	ATL 1113
0203	F	4:00-4:50pm	BPS 1236
ESG1	F	10:50-11:45am	BLD4 4335

Office Hours: Posted on ELMS

Teaching Assistants Sections

James van Doorn	0101
Zoe Lo	0102
Angela Boley (Absences/Extensions)	0103
Derek Miller	0103
Kamran Djourshari	0201
Joanne Jeong	0201
Srikanth Parvathala	0202
Owen Donovan	0203
Scott Mobarrry	ESG1
Ramita Shrestha	ESG1
Syed Addas	
Fahima Chariwala	
Ryan Flint	
Claire Jacobs	
Maksym Kalinin	
Trinity Killip	

Prerequisites

Prerequisite: Minimum grade of C- in INST126

Restriction: Must be in the Information Science program.

Credit granted for: INST327 or BMGT402

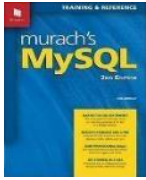
Course Communication

- Use ELMS for messages to instructors and TAs.
- Message subject headings should begin with "INST327 -".
- Project messages should include the team number in the heading.
- Course information will be posted on ELMS. Information will be communicated via the Inbox, Announcements, and Discussion areas. Customize your alerts to ensure you receive notifications via your preferred email account.

Required Resources

Course ELMS Website: <https://umd.instructure.com/courses/1362934>

Textbook: Murach's MySQL, Third Edition (2019), Murach, J. ISBN: 9781943872367



Physical copies are available via campus bookstore.

eBook: <https://www.murach.com/shop/murach-s-mysql-3rd-edition-detail>

Additional readings and resources may be assigned via ELMS.

We will work through SQL examples during lecture. Please have a computer available for course-related work.

Software:

- **MySQL Community Server & Workbench versions 8 or later (alternative XAMPP)**
 - Mac: <https://dev.mysql.com/downloads/mysql/>
NOTE: Specific sub-versions of MySQL are compatible with specific MacOS versions.
 - Windows: <https://dev.mysql.com/downloads/installer/>

Check ELMS for download and installation instructions, videos, and other resources. Feel free to seek help from the instruction team.

Campus Policies

It is our shared responsibility to know and abide by the University of Maryland's policies that relate to all courses, which include topics such as:

- | | |
|---|--|
| ☐ Academic integrity | ☐ Attendance and excused absences |
| ☐ Student and instructor conduct | ☐ Grades and appeals |
| ☐ Accessibility and accommodations | ☐ Copyright and intellectual property |

Please visit www.ugst.umd.edu/courserelatedpolicies.html for the Office of Undergraduate Studies' full list of campus-wide policies and follow up with me if you have questions.

Course Policies

- ☐ Stay engaged with the course and participate.
- ☐ There are many ways you can participate, including completing asynchronous work, attending lecture and discussion sessions, discussions, pair or group work, and project teamwork.
- ☐ Have a computer available for course-related work and bring it to class; (remember to fully charge it).
- ☐ Monitor ELMS daily; adjust settings as needed to ensure receipt of all notifications.
- ☐ Follow course calendar activities:
 - Watch the week's assigned videos, keep up with readings, and execute examples before the lecture.
 - Review practice problems before discussion/lab sessions.
 - Complete worksheets, homework, and quizzes as assigned.
 - Complete lab quizzes during discussion sessions (or soon after) and submit via ELMS.
 - Submit all individual and team assignments via ELMS on time (Some assignments cannot be submitted late, and others have late penalties. If you have an excuse, contact the instructors or TA before the deadline whenever possible!).
 - Do not wait until the deadline to make your ELMS submission! ELMS will mark submissions as "late" even if the submission is only late by 1 or 2 minutes! Technical issues at 10 pm on the due date are NOT our emergency! So, begin assignments early and attend office hours for assistance.
- ☐ All team members **must** participate actively in the team project.
- ☐ If you are struggling with course content, course schedules, please reach out to the instructors and TAs.

Additional University Resources

You are expected to take personal responsibility for your own learning. This includes acknowledging when your performance does not match your goals and doing something about it. Everyone can benefit from some expert guidance on time management, note taking, and exam preparation, so I encourage you to consider visiting <https://tltc.umd.edu/learning-assistance> and scheduling an appointment with an academic coach. Sharpen your communication skills by visiting <https://english.umd.edu/writing-programs/writing-center> and schedule an appointment with the campus Writing Center. Finally, if you just need someone to talk to, visit <http://www.counseling.umd.edu>.

Notice of Mandatory Reporting

Notice of mandatory reporting of sexual assault, sexual harassment, interpersonal violence, and stalking: As faculty members, we are designated as “Responsible University Employees,” and we must report all disclosures of sexual assault, sexual harassment, interpersonal violence, and stalking to UMD’s Title IX Coordinator per University Policy on Sexual Harassment and Other Sexual Misconduct. If you wish to speak with someone confidentially, please contact one of UMD’s confidential resources, such as [CARE to Stop Violence](#) (located on the Ground Floor of the Health Center) at 301-741-3442 or the [Counseling Center](#) (located at the Shoemaker Building) at 301-314-7651. You may also seek assistance or supportive measures from UMD’s Title IX Coordinator, Angela Nastase, by calling 301-405-1142, or emailing titleIXcoordinator@umd.edu. To view further information on the above, please visit the [Office of Civil Rights and Sexual Misconduct's](#) website at ocrsm.umd.edu.

Names/Pronouns and Self Identifications

The University of Maryland recognizes the importance of a diverse student body, and we are committed to fostering equitable classroom environments. We invite you, if you wish, to tell us how you want to be referred to both in terms of your name and your pronouns (he/him, she/her, they/them, etc.). The pronouns someone indicates are not necessarily indicative of their gender identity. Visit trans.umd.edu to learn more.

Additionally, how you identify in terms of your gender, race, class, sexuality, religion, and dis/ability, among all aspects of your identity, is your choice whether to disclose (e.g., should it come up in classroom conversation about our experiences and perspectives) and should be self-identified, not presumed or imposed. We will do our best to address and refer to all students accordingly, and we ask you to do the same for all your fellow Terps.

Office Hours, Topic Clinics, Group Project Support

For extra clarification of a lecture topic, lab exercise, quiz, or assistance with a homework assignment, please attend online or in-person office hours. In addition to the instruction team’s office hours, there will be peer mentor (TAs) office hours that will be announced via ELMS.

Course Structure

The course is delivered in a hybrid learning format structured around weekly modules. As such, there are asynchronous elements and synchronous classes.

- **Self-study Preparation (Asynchronous Materials):** This is a combination of readings, slides, and videos. Students are expected to complete the assigned readings and videos prior to the weekly lecture class time.
- **Lecture and Discussion Classes (Synchronous Sessions):** Lectures and discussion/lab classes will be interactive and based on the material covered in the *Self-Study Preparation*. Participation is necessary to excel at this course. If you anticipate missing a session, email the instructors and TAs ahead of time, or contact us as soon afterwards as possible. At least three "participation instances" (e.g., lecture + lab + one office hour visit or lecture + lab + meeting with your project team) per week are advised and encouraged.

Course Activities and Assessments

This course involves formative and summative assessment components, as well as an applied team project, which serves as the Scholarship in Practice component for the course. Many of the activities and assessments are designed to support the semester-long Team Database Design project. Activities gauge your understanding and expertise of the SQL skill set and readiness to perform the related project work.

- *Discussions (10% of final grade):* Asynchronous online discussions will help with your understanding of relevant concepts in the course. Students will be expected to post a main contribution, responses to other students, and a final contribution summarizing their take-aways from the exchange. [F/I]
- *Lab Exercises (10% of final grade):* Lab exercises provide the skills practice necessary for a comprehensive understanding of SQL and the successful completion of the team project. You will receive these practice problems ahead of time and should preview them before the discussion class, but you should execute them during the discussion class with the TA's assistance as needed. You will submit your work individually via ELMS and will receive a completion grade. Lab exercises will take place during your in-person discussion session and many will be done with your project team to foster team cohesion. On discussion/lab days, attendance is required for full credit on the exercise. Sign-in logs will be used to track attendance. [F/I]
- *Quizzes 10% of the final grade):* Quizzes will check your comprehension of concepts and skills covered in the preceding module. Quizzes are individual work. Most quizzes will become available during the week and will be due on the following Monday. Quizzes will have a 30-minute time limit from when you begin the quiz. [S/I]
- *Homework Assignments (30% of final grade):* There will be several assignments, each of which will include multiple questions. The questions will be practical tasks, such as writing SQL queries, constructing a view, normalizing a table, or developing a stored procedure. The assignments are individual work. Although you may consult with your classmates, TAs, and the instructors to develop general approaches to solving questions, you must work individually while you build, type, test, and debug your answers. Assignment questions will be available on ELMS. Completed assignments will be submitted via ELMS. Timely submission of the completed assignments is essential. The due date of each assignment will be stated explicitly in ELMS. If an assignment due date is a religious holiday for you, please let the instructors know as soon as the assignment is announced so that an alternate due date can be set for you. (See ELMS for due dates). [S/I]
- *Team Project (40% of the final grade):* Students will work in ~5-person teams to design and build a non-trivial relational database. As a Scholarship in Practice course, project-related work is central to this course. The project will involve defining the team structure and responsibilities for individual team members, identifying an information need for a relational database, determining the requirements for the database, developing a deadline-oriented plan and schedule for building the database, designing the logical specifications, building, and populating the database, and developing queries/views that will showcase the capabilities of the database for fulfilling the identified user needs. Students will choose their teams or be assigned by the instructors. Teams will choose their topics from a list of provided datasets. Central to this project will be team management and interaction. Project deliverables will include assessment of team function as well as the software development process. Teaching Assistants (TAs) or Academic Peer Mentors (AMPs) will act as team mentors throughout the life cycle of the project. Teams will provide periodic updates about their project in meetings with their team mentors. [A/T+I]

[F: Formative (20%) - S: Summative (40%) - A: Applied (40%) | I: Individual (62%) - T: Team (38%)]

Grades

Your course grade is determined by your performance on the learning assessments in the course. Assessment scores will be posted on ELMS. If you would like to discuss your grade or have questions about how something was scored, please schedule a time with the course TA. Grade disputes must be submitted in writing within one week of receiving the grade.

A few assignments will allow slightly late submission for a percent deduction on the score. In such cases, the cut-off times and associated penalties will be indicated in the assignment. Assignments submitted beyond the final due date will not be accepted, and you will receive a zero. Not all assignments will allow for late submission. Please read all submission instructions carefully!

This table illustrates the percentage weight of each assessment towards your final grade.

Component	Percentage
Discussions (F/I)	10%
Lab Exercises (F/I)	10%
Quizzes (S/I)	10%
Homework Assignments (S/I)	30%
Team Project (see breakout below) (A)	40%
Team Contract (T)	3%
Proposal (T)	5%
Logical Design (T)	10%
Progress Report with ERD (T)	5%
Peer Assessment and Project Status (I)	2%
Final Project (T)	15%

Letter grades will be assigned using the following scale.

Grading Scale				
A+ ≥ 97%	B+ ≥ 87.00%	C+ ≥ 77.00%	D+ ≥ 67.00%	
A ≥ 93.00%	B ≥ 83.00%	C ≥ 73.00%	D ≥ 63.00%	F <60.0%
A- ≥ 90.00%	B- ≥ 80.00%	C- ≥ 70.00%	D- ≥ 60.00%	

There is no rounding of numeric grades up to the next letter grade.

Course Schedule: (This schedule is for planning purposes and may change. Deadlines are posted in ELMS. Also see ELMS for current information.)

Week/Date	Session	Topics / Videos / Classwork	Reading	Homework	Team Project Assignments
1 1/24 – 1/28	1	Introduction, Syllabus, Course Expectations Database Ethics	Syllabus ELMS		
	2	MySQL Server & MySQL Workbench Installation	Appendices A/B ELMS	MySQL Setup	
2 1/29 – 2/4	3	Relational Database Concepts, SELECT from a single table, Team Project Process & Overview	Chapters 1 & 3 Project Files		
	4	LAB 1: SELECT & MySQL Workbench	Chapter 2	MySQL Setup Complete	Team Formation Survey
3 2/5 – 2/11	5	Relationship types / Joins / UNIONS	Chapter 4		
	6	LAB 2: Multi-table SELECTs		Assignment 1: Basic Queries	Form teams. Students without a team will be assigned to a team.
4 2/12 – 2/18	7	Dependencies, Relationships, Entities/Attributes	Chapter 10		
	8	LAB 3: Modeling			Work with team to assign team roles & responsibilities Team Contract
5 2/19 – 2/25	9	Normalization	Chapter 10		
	10	LAB 4: Normalization		Assignment 2: Multi-Table Queries	Project Proposal
6 2/26 – 3/3	11	Data Types and Functions ERDs, Database Creation, and Indexes	Chapters 8 & 9 Chapters 11		
	12	LAB 5: Datatypes, Functions, & ERD Creation		Assignment 3: Modeling & Normalization	Meet with Mentor to clarify next steps
7 3/4 – 3/10	13	C*UD (INSERT / UPDATE / DELETE)	Chapter 5		
	14	LAB 6: INSERT / UPDATE / DELETE			
8 3/11 – 3/16	15	Aggregate Functions, Summary Queries	Chapter 6		
	16	LAB 7: Summaries Teamwork			
3/17 – 3/24		SPRING BREAK			
9 3/25 – 3/31	17	Subqueries, Common Table Expressions	Chapter 7 ELMS		
	18	LAB 8: Subqueries & CTEs		Assignment 4: C*UD	Project Logical Design

Week/Date	Session	Topics	Reading	Homework	Team Project Assignments
10 4/1 – 4/7	19	Database Schema Creation, Database Backup/Restore			
	20	LAB 9: Forward Engineering / Database Backup		Assignment 5: Aggregation	Meet with Mentor to clarify next steps
11 4/8 – 4/14	21	Views, Stored Programs	Chapters 12, 13, 15		
	22	LAB 10: Views, Stored Programs		Assignment 6: Complex Queries	
12 4/15 – 4/21	23	Stored Programs (Triggers, Events), Concurrency ETL & Sample Data Strategies	Chapters 14, 16		
	24	LAB 11: Data Import		Assignment 7: Views	Project Progress Report
13 4/22 – 4/28	25	Database Administration & Security	Chapters 17 & 18		
	26	LAB 12: Team Project Work			Meet with Mentor to get feedback and plan final steps
14 4/29 – 5/5	27	Workbench Data Import, Database Backup/Restore, and Reverse Engineering	ELMS Chapter 19		
	28	LAB 13: Team Project Work			
15 5/6 – 5/9	29	Teamwork: Final project work			• Project DB & Final Report • Project Reflection

The class meets twice a week for 50 minutes per session unless otherwise noted on page 1 of this syllabus.